1. Lab 11: Link Layer Protocols, Error Detection, and ARP

Lab Name: Lab 11: Link Layer Protocols, Error Detection, and ARP **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

The Link Layer is responsible for transferring datagrams between adjacent nodes across a single physical link or local area network (LAN).

1. **Ethernet II Frame Structure**: Link-layer packets are called **frames**. An Ethernet II frame encapsulates network-layer packets and consists of a header (Destination MAC, Source MAC, EtherType), a payload, and a Frame Check Sequence (FCS) trailing field.
2. **Error Detection**: Ethernet uses a Cyclic Redundancy Check (CRC-32) inside the FCS field. The sender computes a checksum of the frame and appends it to the trailer; the receiver computes the same check to detect bit errors. If an error is detected, the frame is silently discarded.
3. **Address Resolution Protocol (ARP)**: IP addresses are logical and globally unique, but delivery within a local link requires physical MAC addresses. ARP dynamically translates logical 32-bit IPv4 addresses to physical 48-bit MAC addresses. An ARP request is sent as a Layer 2 **broadcast** (destination `ff:ff:ff:ff:ff:ff`), prompting the target host to respond with a unicast ARP reply containing its MAC address.
4. **Layer 2 Switches**: Switches are active link-layer devices that forward frames based on destination MAC addresses. Unlike hubs, switches dynamically build a **MAC address table** (or forwarding database) by inspecting the source MAC address of incoming frames. If a destination MAC is unknown, the switch **floods** the frame out all ports except the source port.

In this lab, you will configure a flat Layer 2 local area network in CML using a L2 switch and three Alpine Linux hosts, perform packet captures of frame structures and ARP resolution, and analyze switch MAC address learning.

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3. Objective / Purpose

To build and configure a flat Layer 2 switched network, analyze the structure of Ethernet II frames, trace the execution of ARP broadcast requests and unicast replies in Wireshark, and verify dynamic MAC address learning in switch forwarding databases.

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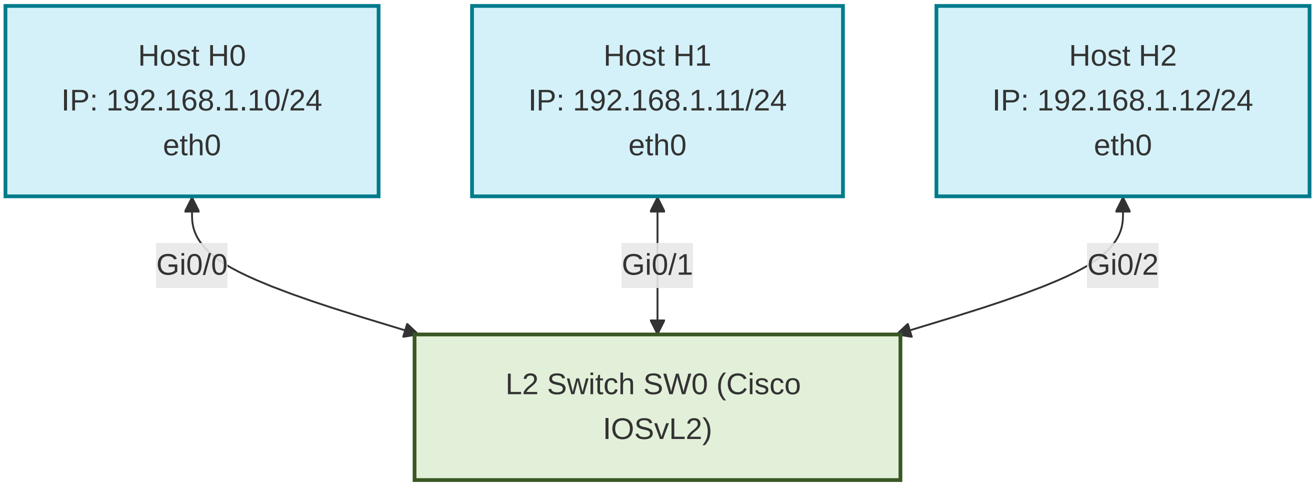
4. Topology Diagram



Details:

- 3x Alpine Linux hosts (`<initials>-H0` , `<initials>-H1` , `<initials>-H2`) connected to a single L2 Switch (`<initials>-SW0`).
- No router is needed since all hosts reside in the same flat IP subnet (`192.168.1.0/24`).

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 3x Linux Nodes, 1x IOSvL2 Switch

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6. Configuration / Methodology

IP Addressing Table

Device	CML Node Name	Interface	IP Address	Subnet Mask	MAC Address
Host 0	`<initials>-H0`	eth0	192.168.1.10	255.255.255.0	Dynamic
Host 1	`<initials>-H1`	eth0	192.168.1.11	255.255.255.0	Dynamic
Host 2	`<initials>-H2`	eth0	192.168.1.12	255.255.255.0	Dynamic
Switch	`<initials>-SW0`	Gi0/0-2	N/A (Switchports)	N/A	Dynamic

Step-by-Step Configuration Guide

1. Configure Host H0 (`<initials>-H0`)

Click on Host H0, open the **Edit Config** tab, and enter:

```
ip addr add 192.168.1.10/24 dev eth0
```

2. Configure Host H1 (`<initials>-H1`)

Click on Host H1, open the **Edit Config** tab, and enter:

```
ip addr add 192.168.1.11/24 dev eth0
```

3. Configure Host H2 (`<initials>-H2`)

Click on Host H2, open the **Edit Config** tab, and enter:

```
ip addr add 192.168.1.12/24 dev eth0
```

4. Configure Switch SW0 (`<initials>-SW0`)

Open the switch's console tab, press **Enter** to get a prompt, and configure the hostname:

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname KH-SW0
KH-SW0(config)# end
KH-SW0# write memory
```

(CML L2 Switch ports are configured as access ports in VLAN 1 by default, requiring no additional port-level configuration).

Step-by-Step Traffic Generation & Execution:

1. **Start Packet Capture**: Right-click the link connecting **Host H0 and Switch SW0**, select **Packet Capture**, and click **Start**.
2. **Collect Local MAC Addresses**: Open the console of H0 and H1 respectively, and run `ip link show dev eth0` (or `ifconfig eth0`) to find their local MAC addresses. Record these for your report.
3. **Trigger ARP and Connectivity**: Open Host H0's console and run:

   ```
   # Clear the ARP cache to force new address resolution pings
   ip neigh flush all

   # Ping Host H1 to trigger ARP and test connectivity
   ping -c 4 192.168.1.11

   # Ping Host H2 to capture another resolution sequence
   ping -c 1 192.168.1.12
   ```

4. **Collect Switch MAC Table**: Open Switch `<initials>-SW0` 's console and run:

   ```
   KH-SW0# show mac address-table
   ```

   Record the output of this table.
5. **Download Capture**: Stop the packet capture in CML and download the `.pcap` capture file to your local computer.

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7. Wireshark Analysis and Questions

Open your downloaded packet capture in Wireshark and answer the following questions:

Part A: Ethernet Frame Analysis

1. Select the first captured **ICMP Echo Request** packet (sent from H0 to H1).
    - What are the source and destination MAC addresses of the Ethernet II frame?
    - Do these match the MAC addresses you recorded from Host H0 and Host H1 via the CLI?
2. What is the value (in hexadecimal) in the **EtherType** field of the Ethernet II frame header? What protocol does this code indicate is encapsulated in the payload?
3. The Ethernet II frame header does not display a Preamble, Start Frame Delimiter (SFD), or Frame Check Sequence (FCS/CRC) in Wireshark.
    - Explain why the network interface card (NIC) and Wireshark typically discard these fields before presenting the packet.
    - Describe the mathematical role of the CRC-32 checksum in error detection.

Part B: Address Resolution Protocol (ARP)

4. Locate the **ARP Request** packet sent from Host H0 before it pinged Host H1.
    - What is the destination MAC address of this frame? Why is this specific MAC address used?
    - Inspect the ARP payload. What is the value of the `Opcode` (operation code) field?
    - What are the values of the sender MAC, sender IP, target MAC, and target IP inside the ARP payload?
5. Locate the corresponding **ARP Reply** packet.
    - What is the destination MAC address of this frame? Is this frame sent as a broadcast or a unicast?
    - What is the value of the `Opcode` field?
    - What critical piece of information did Host H0 learn from this packet?
6. In Host H0's console, run `ip neigh show` (or `arp -a`).
    - What entries are currently stored in H0's ARP table?
    - Explain how caching IP-to-MAC resolutions in the ARP cache prevents hosts from generating broadcast ARP requests for every subsequent packet.

Part C: Switch MAC Address Learning

7. Examine the output of `show mac address-table` on Switch `<initials>-SW0`.
    - List the learned MAC addresses and their associated switch interfaces (ports).
    - Explain the step-by-step logic a switch uses to dynamically learn these MAC addresses as frames flow through it.
    - What action does the switch perform if it receives a frame destined for a MAC address that is *not* currently in its MAC address table?

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Summarize your findings. Explain the roles of MAC addresses in local network framing, how ARP bridges the gap between Layer 3 logical IP addresses and Layer 2 physical MAC addresses, and how L2 switches optimize network performance by segmenting collision domains through dynamic MAC learning.

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10. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- A single PDF report containing:
  - Your written answers to the 7 Wireshark/CLI analysis questions in Section 7.
  - A screenshot of your active CML topology showing the running hosts (`<initials>-H0` , `<initials>-H1` , `<initials>-H2`) and Switch (`<initials>-SW0`).
  - A screenshot of the Switch CLI console displaying the output of `show mac address-table` .
  - An annotated screenshot of your Wireshark capture window showing the ARP Request broadcast and ARP Reply unicast sequence.